

Artificial Intelligence

BCA — Semester 4 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. How do you evaluate any searching algorithm? What are constraint satisfaction problems? Differentiate between Greedy Best First Search and A* search.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. Express the mathematical model of a neuron. Distinguish between learning rule and learning rate. How Back-propagation algorithm is used in learning? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What are the issues in knowledge representation? Given the following knowledge base, using resolution show that "Ava likes watermelon".
a) Ava likes all kind of fruits.
b) Apple and watermelon are fruits.
c) Ava eats watermelon and love it so much.
d) Anything eaten by anybody and love it so much if fruits.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Explain the different steps in NLP.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Describe the components of expert system.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is the role of rewards and punishment in reinforcement learning? Describe with an example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Represent the following knowledge using conceptual dependency.
a) He opened the door.
b) Ram gave rose to Sita.
c) She is crying.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Distinguish between Depth First Search and Breadth First Search.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. What is rational agent? List some applications of AI.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain about unification and lifting with example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Distinguish between simple reflex agent and model based agent.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. How does script can be used in representing knowledge? Explain.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.